Resume of Mingwu ZHANG

Basic Information



School :	School of Computer Science
Gender：	Male
Date of Birth：	
Title：	Professor
Education：	Ph.D of Engineering
Tutor：	Phd and Master degree
Email:	scauzhang@gmail.com
Interest of research:	Information Security, AI, Big-data

Visiting Experience

2010/08-2012/09, JSPS Fellow/ Postdoctoral, Kyushu University, Japan;
2015/05-2016/06, Visiting scholar, University of Wollongong, Australia;

Representative Projects

[1] JSPS, "*Privacy-carrying Cryptography and its Applications in Decentralized Trust Systems*", Japan, Project leader.

[2] NSFC, "*Research on Witness Encryption based on Relationship Language on NP Complete*", China, Project leader.

[3] NSFC, "*Crowdsensing Security based on Blockchain*", Adjunct Project leader.

[4] NSFC, "*Key Techniques for Provably Secure Program Obfuscations*". China, Project leader.

[5] NSFC, "*Research on Leakage-Resilient Generalized Identity-based Cryptography and its Provable Security*". China, Project leader.

[6] NSFC, "*Functional Encryption Schemes with Privacy Preservation*". China, Project leader.

[7] Key Research and Development Program of Hubei Province, "*Monitoring Intelligence Analysis in Blockchain Security*", China, Project leader.

[8] "*Detection and Analysis for Malicious Network Traffic*", China, Project leader.

[9] Natural Science Foundation of Hubei Province, "*Privacy Preservation in Present of Information Protection*", China, Project leader.

[10] Natural Science Foundation of Guangdong Province, "*Research on Reputation Model of Internet-of-Things with Privacy Protection*" , China, Project leader.

Representative Articles (More refer DBLP and Google Scholar)

[1] **Mingwu Zhang**, Shijin Chen, Jian Shen and Willy Susilo. "PrivacyEAFL: privacy-enhanced aggregation for federated learning in mobile crowdsensing". *IEEE Transactions on Information Forensics and Security*, 18:5804-5816 (2023)

[2] **Mingwu Zhang**, Yu Chen and Willy Susilo. "Decision tree evaluation on sensitive datasets for secure e-healthcare systems". *IEEE Transactions on Dependable and Secure Computing*, 20(5): 3988-4001(2023)

[3] Hua Shen, Ge Wu, Zhe Xia, Willy Susilo, **Mingwu Zhang***. "A privacy-preserving and verifiable statistical analysis scheme for an e-commerce platform". *IEEE Transaction on Information Forensics and Security* 18:

2637-2652,2023

[4] **Mingwu Zhang**, Mingxuan Yang, Gang Shen, Zhe Xia, Yuntao Wang. “A verifiable and privacy-preserving cloud mining pool selection scheme in blockchain of things”. *Information Sciences*, 623 (2023) 293-310 (2023)

[5] Guoqiang Deng, Min Tang, Yuxing Xi, **Mingwu Zhang***. “Privacy-preserving online medical prediagnosis training model based on soft-margin SVM”. *IEEE Transaction on Service Computing*, 16(3): 2072-2084 (2023)

[6] Biwen Chen, Tao Xiang, Guoli Xiao, **Mingwu Zhang**, Debiao He, Yan Zhang. “Efficient attribute-based signature with collusion resistance for Internet of vehicles”. *IEEEE Transactions on Vehicular Technology*, 72(6): 7844-7856 (2023)

[7] Hua Shen, Jiqiang Li, Ge Wu and **Mingwu Zhang***, “Data release for machine learning via correlated differential privacy”. *Information Processing & Management,* 60(3): 103349 ,2023

[8] **Mingwu Zhang**, Mingxuan Yang, Gang Shen. “SSBAS-FA: A secure sealed-bid e-auction scheme with fair arbitration based on time-released blockchain”. *Journal of System Architecture*. 129:102619 (2022)

[9] **Mingwu Zhang**, Yu Chen and Jingkang Lin. “A privacy-preserving optimization of neighborhood-based recommendation for medical-aided diagnosis and treatment”. *IEEE Internet of Things Journal.* 8(13)**:** 10830-10842 (2021)

[10] Hua Shen, **Mingwu Zhang***, Jian Shen. “Efficient privacy-preserving cube-data aggregation scheme for smart grids”. *IEEE Transaction on Information Forensics and Security*, 12(6):1369-1381 (2017)

[11] Yanwei Zhou, Bo Yang, Zirui Qiao, Zhe Xia, **Mingwu Zhang** and Yi Mu. “Leakage-resilient identity-based cryptography from minimal assumption”. *Des. Codes Cryptography*, 91, 3757-3801 (2023)

[12] **Mingwu Zhang**, Shuo Huang, Gang Shen, Yuntao Wang. “PPNNP: A privacy-preserving neural network prediction with separated data providers ssing multi-client inner-product encryption”. *Computer Standard and Interfaces*, 84:103678 (2023)

[13] Xingxin Li, Youwen Zhu, Jian Wang, Zhe Liu, Yining Liu, **Mingwu Zhang**. “On the soundness and security of privacy-preserving SVM for outsourcing data classification”. *IEEE Transactions on Dependable and Secure Computing (IEEE-TDSC)*, 15(5): 906-912, 2018

[14] Run-Hua Shi, Bai Liu, **Mingwu Zhang**. Measurement-device-independent quantum secure multiparty summation. *Quantum Information Processing*. 21(4): 122 (2022)

[15] Yanwei Zhou, Bo Yang, Zhe Xia, **Mingwu Zhang**, Yi Mu. “Novel generic construction of leakage-resilient PKE scheme with CCA security”. *Des. Codes Cryptography*. 89(7): 1575-1614 (2021)

[16] **Mingwu Zhang**, Yu Chen and Jingkang Lin. “A privacy-preserving optimization of neighborhood-based recommendation for medical-aided diagnosis and treatment”. *IEEE Internet of Things Journal*. 8(13)**:** 10830-10842 (2021)

[17] Yanwei Zhou, Yuan Xua, Zirui Qiao, Bo Yang, **Mingwu Zhang**. “Continuous leakage-resilient certificate-based signcryption scheme and application in cloud computing”. *Theoretical Computer Science*, 860: 1-22 (2021)

[18] **Mingwu Zhang**, Yu Chen, Willy Susilo. “PPO-CPQ: A privacy-preserving optimization of clinical pathway query for e-healthcare systems”. *IEEE Internet of Things Journal*, 7(10): 10660-10672 (2020)

[19] **Mingwu Zhang**, Yan Jiang, Hua Shen, and Susilo Willy. “Cloud-based data-sharing scheme using verifiable and cca-secure re-encryption from indistinguishability obfuscation”. *Inscrypt’18*, LNCS11449, pp240-259, Springer (2019)

[20] Rongmao Chen, Yi Mu, Guomin Yang, Willy Susilo, Fuchun Guo, **Mingwu Zhang**. “Cryptographic reverse firewall via malleable smooth projective hash function”. *ASIACRYPT 2016*, Part I, LNCS 10031, pp 844-876 (2016)